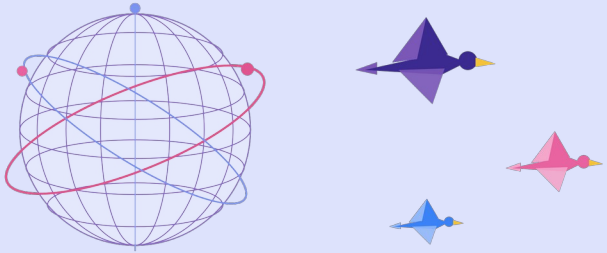
PARTICIPANT PRESENTATION

Qiskit Fall Fest 2026

Hackathon · Team Details



Participant Details

PROJECT TITLE
Nirikshak: Drift-Robust Eavesdropper Detection for BB84 QKD

TRACK
Track 1: Quantum Cryptography & Security

TEAM NAME
Essential Engineers · HC-0173

TEAM LEADER
J Bhuvanesh

EMAIL / CONTACT
github.com/Bhuvan-esh

TEAM MEMBERS
J Bhuvanesh (Lead) · Dipankar Chettri · Anusha A

INSTITUTION / ORGANISATION
Shridevi Institute of Engineering and Technology

COUNTRY
India

Problem Statement

Eavesdropper detectors for QKD are trained and tested almost entirely in simulation. Nobody has measured whether they can tell normal hardware drift from a real attack on real quantum hardware, so deployed QKD links risk false alarms that discard valid keys, or eavesdroppers who go undetected.

WHY IT MATTERS

- **The 11% rule is not an attack detector.** It is the security proof's limit for distilling any secret key (Shor-Preskill). Intercepting 10% of rounds adds only about 2.5% QBER, so the operator never learns the link is tapped, cannot locate it, and cannot tell an attack from drift.
- **Real devices drift.** Temperature, aging and recalibration shift the noise; simulation-trained QKD detectors degrade sharply under shifted conditions (arXiv 2610.25203, arXiv 2605.27497).
- **India's QKD rollout is accelerating:** a 1,000 km National Quantum Mission network, C-DOT QKD products, DRDO field and live-network trials, while ANSSI, BSI, NLNCSA and Swedish NCSA say QKD is not yet mature from a security perspective.

Objectives & Novelty

OBJECTIVES

- Run honest BB84 sessions and qubit-level attacks on IBM Quantum hardware.
- Stress-test simulation-trained detectors (11% rule, sim-calibrated threshold, Random Forest, XGBoost) on hardware data.
- Build a detector whose false-alarm rate is stated in advance, even under device drift.
- Release an open dataset of hardware BB84 attack logs, a reusable scorecard and a live dashboard.

NOVELTY

- **Hardware-in-the-loop:** attacks run as real qubit operations across qubits, 22 hours and three IBM machines.
- **Per-session calibration:** an honest reference circuit on the same qubits sets today's threshold; a permutation test gives one verdict and p-value.
- **Fair emulation:** same timing and spin echo on idle qubits, so only Eve differs.
- **Honest scope:** optical-only attacks (photon-number splitting, blinding) are simulated and labelled.

Solution & Quantum Advantage

We run honest BB84 sessions and qubit-level attacks on IBM Quantum hardware, stress-test simulation-trained detectors on that data, and build a detector that stays calibrated under device drift.

QUANTUM ADVANTAGE

BB84's security is physical: an eavesdropper's measurement disturbs the qubits and shows up as errors (QBER). Nirikshak turns that disturbance into a calibrated, per-session alarm on real quantum hardware.

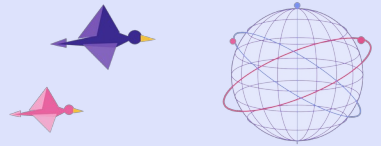
KEY IDEA

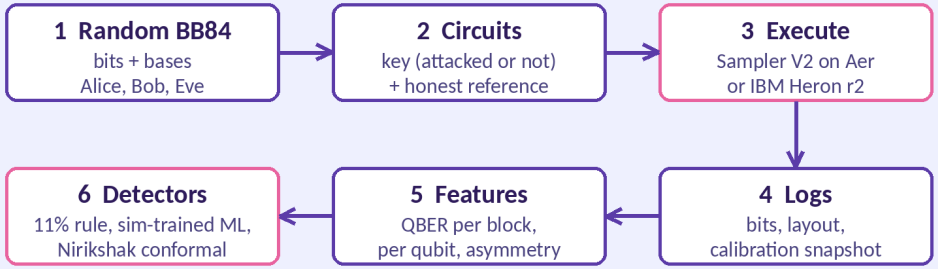
A separate honest calibration circuit on the same qubits (no channel, so Eve cannot touch it) measures today's noise. Split-conformal calibration sets the alarm threshold, so the false-alarm rate is stated in advance, drift or not, and small attacks are flagged far below 11%.

WHY THIS WORKS

- **Real attacks on real qubits:** intercept-resend (full/partial) and an entangling probe, repeated across qubits and time.
- **Sim-to-hardware stress test:** sim-calibrated QBER threshold, Random Forest and XGBoost tested on drifted simulation, a Kingston snapshot and hardware.
- **Drift-robust detector:** each session's reference blocks set its own threshold.
- **Fair emulation:** matched timing plus spin echo, so only Eve differs.
- **Honest scope:** optical-only attacks are simulated and labelled.

Algorithm & Circuit Design





Attacks on real qubits (between preparation and measurement)

- Intercept-resend: mid-circuit measurement in a random basis (1, 2, 4 or 8 of 16 rounds)
- Entangling probe: controlled rotation onto Eve's ancilla qubit

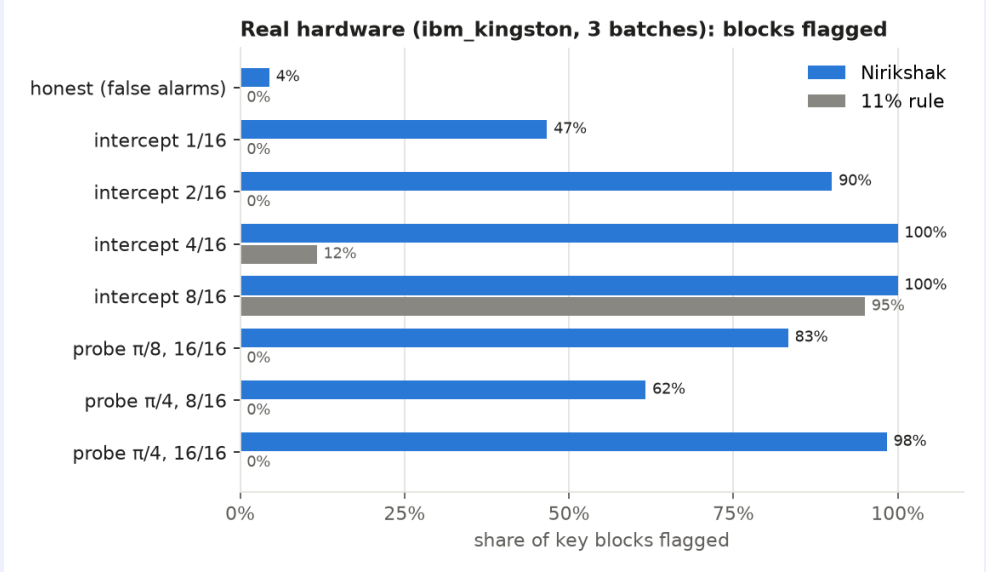
Nirikshak: honest reference circuit on the same qubits sets today's threshold
(split-conformal) and a permutation test gives one verdict + p-value per session.

- **Input:** random bits and bases, 16 rounds per circuit (one per qubit), swept as Sampler V2 parameter bindings.
- **Core:** BB84 prepare-and-measure circuits with intercept-resend (random-basis mid-circuit measurement) or an entangling probe (controlled rotation onto Eve's ancilla).
- **Execution:** same Sampler V2 interface on Aer and IBM hardware; one job = 10 sessions (key + reference circuit); qubits chosen from live calibration.
- **Post-processing:** sifting, QBER per basis, basis asymmetry, per-qubit error rates.
- **Output:** per-session verdict and p-value, block alarms, BB84 key rates, Streamlit dashboard.

Implementation & Tech Stack

Language	Python 3.11+ (developed on 3.12)
Quantum SDK	Qiskit 2.6.0, Qiskit IBM Runtime 0.50.0 (Sampler V2, job mode)
Simulation	Qiskit Aer 0.18.0 (matrix product state), FakeKingston noise
Hardware	IBM Quantum Open Plan: ibm_kingston, ibm_marrakesh, ibm_fez (Heron r2)
ML	scikit-learn 1.9.1, XGBoost 3.4.1
Detectors	11% QBER rule, sim-calibrated threshold, Random Forest, XGBoost, Nirikshak (per-session conformal)
Statistics	Split-conformal thresholds, permutation test (session verdict), Clopper-Pearson intervals (SciPy)
Frontend	React 19 + TypeScript (Vite), hand-built SVG charts, GitHub Pages; Streamlit 1.66.0 dashboard; matplotlib figures
Testing	pytest, GitHub Actions (tests.yml)
Version control	Git + GitHub

Results & Analysis



- **ibm_kingston, 3 batches over 22 h:** all 21 attacked sessions flagged ($p \leq 0.0002$), 0 of 9 honest.
- **Smallest attack (1 of 16 rounds):** QBER only 3.0-3.2%; Nirikshak flags 47% of blocks and every session, the 11% rule 0%.
- **ibm_marrakesh (3.6% noise) and ibm_fez (2.0%):** 7 of 7 attacked sessions flagged on each, nothing retuned.
- **All three machines:** 35 of 35 attacked, 0 of 15 honest flagged.
- **Under noise shift,** sim-trained detectors flag up to 88% of honest blocks; Nirikshak's worst case is 13%.

Impact & Real-World Applications

THE GAP IT FILLS

BB84 can keep producing keys while a partial eavesdropper listens. On ibm_kingston, every attack except 8 of 16 rounds still leaves a positive key rate (0.16-0.61).

Nirikshak is the missing intrusion-detection layer: the operator learns the link is tapped and can tell an attack from drift.

NETWORK MONITOR PROTOTYPE

Runs on a stream of per-block link telemetry (link, time, key or reference, sifted bits, errors). No key material is needed.

- Honest windows flagged: 5.3% (2,100 windows)
- Link tapped on 1 of 16 rounds: 100% flagged after the tap; 11% rule 0
- Drifting link: 1.0 false block alarm vs 8.2 for a frozen threshold

CONTEXT AND OUTPUTS

Relevant to India's 1,000 km National Quantum Mission QKD network, C-DOT QKD products and DRDO trials.

- Open dataset of hardware BB84 attack logs
- Reusable detector scorecard
- Live dashboard: Streamlit + React web app

Challenges & Future Scope

CHALLENGES

- **Timing artifact:** batch 1 qubits idled during Eve's measurements, inflating detection to 100%. Excluded; fixed with matched timing.
- **Idle dephasing:** honest QBER 4.2% (X basis 7%) until a spin echo brought it to 1.7%.
- **Qubit-0 readout failure** not predicted by IBM calibration, so qubits are chosen from live calibration.
- **Budget:** 600 s quantum time per 28 days, at most 3 queued jobs (73 s used).

LIMITATIONS AND FUTURE SCOPE

- **More hardware data:** 15 honest sessions bound the session false-alarm rate only to 0-22%; the 5% claim rests on 600 simulated sessions.
- **Photonic realism:** decoy states, detector statistics, finite-key effects; optical attacks are not yet run on hardware.
- **Deployment:** operator integration for the network monitor, validated beyond simulated binomial links.
- **Tighter block alarms:** vary up to 15% in one honest session since thresholds use 20 reference blocks.

CLOSING

Thank You

Questions?

Inspiring Minds. Shaping Tomorrow.

GITHUB REPOSITORY
github.com/vtu-qiskit-fall-fest-2026/vtu-hc-essential-engineers

DEMO VIDEO LINK
youtu.be/gO9v1e-yHNs

